Satria Hadi Wiyana

Machine Learning Engineer & Full-Stack Developer | satriahadiwiyana7@gmail.com | linkedin.com/in/satriahadiwiyana

Machine Learning Engineer and Full-Stack Developer driven by a vision to build highly efficient and impactful intelligent software. Specialized in AI architectures powered by NLP and Computer Vision, with a strong emphasis on local model inferencing to ensure data privacy and cost efficiency. Experienced in architecting end-to-end application ecosystems, from database optimization to production deployment. Beyond writing code, I prioritize clear communication—distilling high-level computational complexity into concise, actionable, and practical technical insights. I operate as an adaptive collaborative partner, resolving logical bottlenecks and driving product scalability in fast-paced environments.

WORK EXPERIENCE

Forum Assisten - AMIKOM Yogyakarta 2024 – 2026

Teaching Assistant

Mentored over 50 students per semester in core computer science practical courses, including Algorithms & Programming, Operating Systems, and Computer Networks. Responsible for translating and streamlining complex technical modules from lecturers into concise and easily digestible materials. By utilizing a peer-to-peer mentoring approach and positioning myself as a learning partner, effectively boosted student engagement and active participation during practical sessions. Furthermore, routinely conducted personalized code reviews and provided troubleshooting assistance to ensure they thoroughly understood the practical implementation of computational logic.

Web Master - The Nganjuk Post (TNP) 2020 – 2023

Web Master

Managed the end-to-end operations and architecture of The Nganjuk Post (TNP) digital news portal using WordPress CMS. Fully responsible for ensuring server stability and uptime to reliably serve an active traffic of over 2,000 monthly readers. Primary focus included optimizing page load speeds and bandwidth efficiency by implementing caching mechanisms and image compression, effectively delivering a lightweight and highly responsive portal.

RELEVANT PROJECTS

TryLab - Synthetic Mobile UI/UX Testing 2026 - Present

AI Engineer

Spearheaded and developed an autonomous AI agent for automating UI/UX usability testing on mobile applications. By integrating local Vision-Language Models (VLMs) such as Llama 3.2 Vision and Qwen-VL, the agent interprets visual interface elements using Set-of-Marks strategies, navigates autonomously via ADB commands, and automatically generates persona-based user interaction reports without human intervention.

LiveGuard AI - Face Anti-Spoofing Detection System 2026

Machine Learning Engineer

Developed a face liveness detection system for the Find IT competition utilizing a cutting-edge ensemble architecture based on EfficientNet-B3 and CDCN. The experiment focused on enhancing model robustness through pseudo-labeling techniques and image preprocessing optimization. Systemically, designed a data flow logic allowing anonymous users to perform direct inference, with strict access separation ensuring that historical logs are strictly accessible only to authenticated users.

Aegis NEO - Orbital Threat Intelligence Analytics 2025

Data Scientist

Built a Big Data analytics system to predict the hazard levels of asteroid trajectories. Utilized the PySpark framework and Hadoop ecosystem to handle large-scale historical data processing, while implementing CatBoost and XGBoost algorithms to deliver high-performance threat classification for Near-Earth Objects (NEOs) with precise evaluation metrics.

AgriRAG - Plant Disease & Fertilizer Recommendation Engine 2025

AI Developer

Developed an intelligent Retrieval-Augmented Generation (RAG) system to diagnose plant diseases and provide precise fertilizer recommendations based on agricultural guidelines. Integrated the LangChain framework, embedding architectures, and a vector database to ensure highly relevant context retrieval, leveraging a local LLM to generate accurate and actionable guidance for farmers.

CekKontrak - MSME Legal Document NLP Analysis 2025

NLP Engineer

Designed and trained Natural Language Processing (NLP) models to extract crucial information and analyze clauses within cooperative agreement documents. Built this system to automate legal contract reviews for MSMEs, detect text anomalies, and minimize business legal risks by significantly accelerating the document validation process.

SIMBATRAS - Integrated Waste Bank Management Platform 2023 - 2024

Full-Stack Developer

Built an end-to-end integrated waste bank management platform utilizing the Laravel (PHP) framework for the backend and React (JavaScript) for the frontend. Designed the relational database architecture and developed a dynamic dashboard to track customer balances and calculate financial reports. In the production phase, I implemented efficient record update strategies to maintain data integrity and system stability for the platform successfully deployed to production (simbatras.com).

EDUCATION

Universitas Amikom - Yogyakarta 2023 - Present

Bachelor of Computer Science in Informatics , GPA: 3.76/4.00

Focused on artificial intelligence, machine learning algorithms, and software engineering. Actively translating computational theory into real-world project implementations and large-scale data processing research.

CERTIFICATIONS & TRAINING

DBS Foundation x Dicoding Coding Camp 2026 - Present

Scholarship Recipient in - AI Engineer Learning Path

An industry-standard intensive training program via a full scholarship path. Focused on designing, training, and optimizing Machine Learning and Deep Learning architectures, while validating technical capabilities through real-world case studies focused on model efficiency and scalable AI deployment.

TECHNICAL SKILLS

- **Machine Learning & Artificial Intelligence (AI)** : Natural Language Processing (NLP), Computer Vision, Retrieval-Augmented Generation (RAG), Vision-Language Models (VLM), Ensemble Methods.
- **Data Science & Big Data** : PySpark, Hadoop, CatBoost, XGBoost, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, OpenCV.
- **Software & Web Development** : Python, PHP, JavaScript, C, C#, React.js, Laravel, Tailwind CSS, Flask.
- **Infrastructure & Environment** : Local LLM/VLM Inferencing (Llama 3.2, Qwen-VL), Git, WSL2 (Local Deep Learning Environment), MySQL, RESTful API Integration.